

1. What is Regression?

Answer

Regression is a **supervised machine learning technique** used to predict a **continuous numerical value**.

Regression and classification are both supervised machine learning techniques, but they differ mainly in the type of output they predict. Regression is used when the target variable is a continuous numerical value, such as predicting a house price of ₹75,00,000, a salary of ₹60,000, or sales revenue of ₹10,00,000. Classification, on the other hand, is used when the target variable belongs to a specific category or class. For example, a classification model may predict whether a house is “Affordable,” “Medium-priced,” or “Expensive.” Similarly, it may predict whether a customer will default on a loan or not, or whether a transaction is fraudulent or genuine. In a house price prediction project, if the target is the exact numerical price, the problem is regression. However, if the target is a category such as “Low Price,” “Medium Price,” or “High Price,” the problem becomes classification. Therefore, the primary difference is that regression predicts numerical values, while classification predicts predefined categories or labels.

Explanation

In regression, the machine learning model learns the relationship between one or more input features and a numerical target variable. After learning these relationships from historical data, the model predicts a new numerical value for unseen data.

For example, in a house price prediction project, the model learns how features such as **area, bedrooms, bathrooms, location score, and property age** affect the final house price.

Example

If we provide:

- Area = 2,000 sq. ft.
- Bedrooms = 3
- Bathrooms = 2
- Location Score = 8

The model may predict:

House Price = ₹75,00,000

Use in a Real ML Project

In a **House Price Prediction project**, regression is used because `house_price_inr` is a continuous numerical target variable.

Input Features



Area, Bedrooms, Bathrooms, Location



Regression Model



Predicted House Price

₹75,00,000

2. What is the Difference Between Regression and Classification?

Answer

Simple Linear Regression is a supervised machine learning algorithm that uses one independent variable to predict one continuous dependent variable. It assumes that there is a linear relationship between the input variable and the target variable. The objective of the algorithm is to find the best-fitting straight line through the data points so that the difference between the actual values and predicted values is minimized. In a house price prediction project, Simple Linear Regression can use only one feature, such as `area_sqft`, to predict `house_price_inr`. The model tries to understand whether the house price increases or decreases as the area changes. The mathematical formula is $y = \beta_0 + \beta_1 x$, where y represents the predicted target value, x represents the independent variable, β_0 represents the intercept, and β_1 represents the coefficient or slope. The coefficient indicates how much the predicted target changes when the input feature changes by one unit. Simple Linear Regression is easy to understand and is often used as a baseline model, but it may not be sufficient for real-world problems where multiple factors influence the target variable.

Regression predicts continuous numerical values, whereas classification predicts categories or labels.

Explanation

The main difference is the type of output produced by the machine learning model. Regression produces a number that can have many possible values, while classification assigns data to a specific category.

Example

Regression

Predicting house price:

Input → House Features

Output → ₹75,00,000

Classification

Predicting house category:

Input → House Features

Output → Affordable / Medium / Expensive

Real-World Examples

Regression

Predicting house price

Predicting salary

Predicting sales revenue

Predicting temperature

Classification

Predicting loan default

Predicting spam or not spam

Predicting fraud or genuine

Predicting disease or no disease

Use in a Real ML Project:

In a **House Price Prediction project**, the problem is regression because:

house_price_inr = ₹52,50,000

is a continuous numerical value.

If the target were:

Affordable

Medium

Expensive

then it would become a classification problem.

3. What is Simple Linear Regression?

Answer

Simple Linear Regression is a regression algorithm that uses **one independent variable** to predict **one continuous dependent variable**.

Explanation

The model tries to find the best straight-line relationship between one input feature and the target variable.

For example, if we use only area_sqft to predict house_price_inr, the model tries to understand how the price changes when the area increases.

Mathematical Formula

$$y = \beta_0 + \beta_1 x$$

Where:

- y = predicted value
- x = input feature
- β_0 = intercept

- β_1 = coefficient

Example

Suppose:

$$\text{House Price} = 10,00,000 + 5,000 \times \text{Area}$$

For a house with an area of 1,000 sq. ft.:

$$\text{House Price} = 10,00,000 + (5,000 \times 1,000)$$

The model predicts the house price based only on the area.

Use in a Real ML Project

Simple Linear Regression can be used as a **baseline model** in a house price prediction project.

area_sqft



Linear Regression



house_price_inr

However, real house prices depend on many factors, so using only area may not provide the most accurate predictions.

4. What is Multiple Linear Regression?

Answer

Multiple Linear Regression uses **multiple independent variables** to predict one continuous target variable.

Explanation

House prices are affected by multiple factors. For example, a larger house may cost more, but location, number of bedrooms, bathrooms, age of the property, and amenities also influence the price.

Multiple Linear Regression considers all these features together to make a prediction.

Mathematical Formula

$$\text{Price} = \beta_0 + \beta_1 \text{Area} + \beta_2 \text{Bedrooms} + \beta_3 \text{Bathrooms} + \beta_4 \text{Age} + \epsilon$$

Where:

- β_0 = intercept

- $\beta_1, \beta_2, \beta_3$ = coefficients
- *Area, Bedrooms, Bathrooms, Age* = input features
- ϵ = error

Example

The model may use:

area_sqft = 2000

bedrooms = 3

bathrooms = 2

property_age = 5

location_score = 8

to predict:

house_price_inr = ₹80,00,000

Use in a Real ML Project

Multiple Linear Regression is useful when your dataset contains several numerical features related to the target.

For example:

```
features = [
    "area_sqft",
    "bedrooms",
    "bathrooms",
    "property_age",
    "location_score"
]
```

target = "house_price_inr"

The model learns the combined influence of all features.

5. What is Regularized Regression?

Answer

Regularization is a technique used to **reduce overfitting** by adding a penalty to large model coefficients.

Explanation

Sometimes a regression model becomes too complex and learns unnecessary patterns from the training data. This can cause excellent performance on training data but poor performance on new data.

Regularization controls the model by penalizing large coefficients.

The two important regularized regression algorithms are:

1. **Ridge Regression**
 2. **Lasso Regression**
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Ridge Regression

Answer

Ridge Regression uses **L2 regularization**.

Formula

$$Loss = MSE + \lambda \sum \beta^2$$

The penalty is based on the **square of the coefficients**.

Example

If a feature has a very large coefficient, Ridge Regression reduces its size.

Before Regularization:

Feature coefficient = 100

After Ridge:

Feature coefficient = 65

Real ML Project Use

Ridge Regression is useful when:

- Many features are important
 - Features are correlated
 - You want to reduce overfitting
 - You do not want features to be completely removed
-

Lasso Regression

Answer

Lasso Regression uses **L1 regularization**.

Formula

$$Loss = MSE + \lambda \sum |\beta|$$

Lasso can reduce some coefficients completely to zero.

Example

Before Lasso:

area_sqft = 0.80
bedrooms = 0.40
crime_rate = 0.05
unused_feature = 0.02

After Lasso:

area_sqft = 0.75
bedrooms = 0.35
crime_rate = 0
unused_feature = 0

Real ML Project Use

Lasso can perform **automatic feature selection**.

In a house price project, if some features contribute very little to price prediction, Lasso may assign them a coefficient of zero.

6. What is Overfitting?

Answer

Overfitting occurs when a machine learning model learns the training data too closely, including noise and unnecessary patterns.

Explanation

An overfitted model memorizes the training data instead of learning general patterns. As a result, it performs very well on training data but performs poorly on new, unseen data.

Example

Training R² Score = 0.99

Testing R² Score = 0.55

This large difference may indicate overfitting.

Real ML Project Example

Suppose a house price model memorizes specific houses from the training dataset.

It may predict training houses accurately:

Actual: ₹80,00,000

Predicted: ₹79,95,000

But for a new house:

Actual: ₹70,00,000

Predicted: ₹95,00,000

The model has not learned general patterns properly.

How to Reduce Overfitting

- Use Ridge or Lasso Regression
 - Reduce unnecessary features
 - Use cross-validation
 - Use simpler models
 - Collect more data
 - Tune model hyperparameters
-

7. What is Underfitting?

Answer

Underfitting occurs when a model is too simple to learn the important patterns in the data.

Explanation

An underfitted model performs poorly on both training and testing data because it has not learned enough from the dataset.

Example

Training R^2 Score = 0.45

Testing R^2 Score = 0.40

The model is performing poorly on both datasets.

Real ML Project Example

Suppose we predict house prices using only:

Area

But actual prices also depend on:

- Location
- Bedrooms
- Bathrooms
- Property age
- Garage
- Pool

The model may be too simple and underfit the data.

8. What is MAE?

Answer

MAE, or **Mean Absolute Error**, measures the average absolute difference between actual and predicted values.

Formula

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of observations

Example

Suppose the prediction errors are:

₹1,00,000

₹2,00,000

₹3,00,000

The average error is:

MAE = ₹2,00,000

Human-Writable Interpretation

If the MAE of a house price prediction model is:

₹5,00,000

it means that the model's predictions are wrong by approximately **₹5 lakh on average**.

Use in a Real ML Project

MAE is useful when you want an easy-to-understand error value in the same unit as the target.

MAE = ₹3,50,000

This is easy to explain to business stakeholders.

9. What is RMSE?

Answer

RMSE, or **Root Mean Squared Error**, measures prediction error and gives greater importance to large errors.

Formula

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

Explanation

RMSE squares the errors before calculating the average. Therefore, large errors become much more significant.

Example

Actual price:

₹1,00,00,000

Predicted price:

₹50,00,000

The error is:

₹50,00,000

Because this is a very large error, RMSE penalizes it heavily.

Use in a Real ML Project

RMSE is especially useful for house price prediction because a very large prediction error can have serious financial consequences.

For example:

Model A RMSE = ₹8,00,000

Model B RMSE = ₹4,00,000

Model B generally makes smaller prediction errors.

10. What is R² Score?

Answer

R^2 Score measures how much variation in the target variable is explained by the machine learning model.

Explanation

R^2 tells us how well the model explains the changes in house prices based on the input features.

Example

Suppose:

R^2 Score = 0.85

This means the model explains approximately **85% of the variation in house prices** using the available features.

The remaining 15% may be caused by:

- Unavailable features
- Market changes
- Data noise
- Other unknown factors

General Interpretation

R^2 Score Interpretation

1.0	Perfect prediction
0.90	Excellent
0.80	Very good
0.60	Moderate
0.30	Weak
0	Model explains no variation

Use in a Real ML Project

R^2 Score = 0.88

This means the house price model explains approximately **88% of the variation in house prices** in the dataset.

11. What is Cross-Validation?

Answer

Cross-validation is a model evaluation technique used to evaluate how well a machine learning model performs on different subsets of data.

Explanation

Instead of dividing the dataset only once into training and testing data, cross-validation divides the training data into multiple parts called **folds**.

In **K-Fold Cross-Validation**, the dataset is divided into K parts.

For example, in 5-Fold Cross-Validation:

Fold 1 → Testing

Fold 2 → Testing

Fold 3 → Testing

Fold 4 → Testing

Fold 5 → Testing

The model is trained and evaluated five times.

Example

Suppose the results are:

Fold 1 = 0.82

Fold 2 = 0.85

Fold 3 = 0.81

Fold 4 = 0.84

Fold 5 = 0.83

Average R^2 :

Average $R^2 = 0.83$

Use in a Real ML Project

Cross-validation helps determine whether a model performs consistently.

For example:

Random Forest:

CV R^2 Score = 0.86 ± 0.03

This indicates that the model is performing consistently across different data subsets.

Complete Regression Workflow in a Real ML Project

For your **House Price Prediction Project**, the complete process can be:

1. Collect House Price Dataset



2. Understand the Data



3. Clean Missing Values



4. Perform EDA



5. Select Features



6. Split Data

X = Input Features

y = house_price_inr



7. Train Regression Models



Linear Regression

Ridge Regression

Lasso Regression

Decision Tree

Random Forest



8. Evaluate Models



MAE

RMSE

R² Score



9. Cross-Validation



10. Hyperparameter Tuning



11. Select Best Model



12. Predict New House Prices

Final Project Example

Area = 2,000 sq. ft.

Bedrooms = 3

Bathrooms = 2

Location Score = 8

Property Age = 5 years

Model Prediction

Predicted House Price = ₹78,50,000

Model Evaluation

MAE = ₹3,00,000

RMSE = ₹4,50,000

$R^2 = 0.87$

The regression model predicts house prices using multiple property features. An R^2 score of 0.87 indicates that the model explains approximately 87% of the variation in house prices. The MAE of ₹3 lakh means that the model's predictions are wrong by approximately ₹3 lakh on average, while the RMSE of ₹4.5 lakh indicates that some larger prediction errors exist.